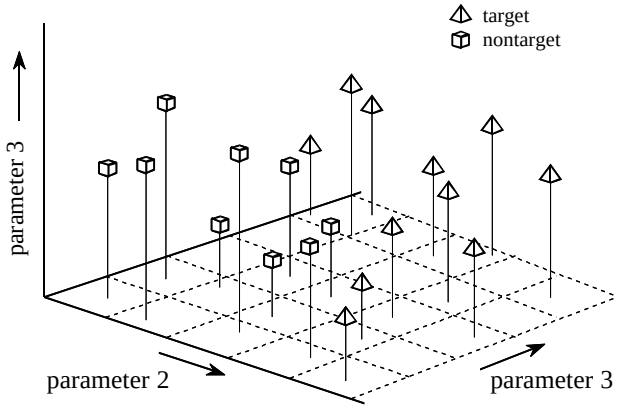

FIGURE 26-4
Example of a three-parameter space. Just as a two-parameter space forms a plane surface, a three parameter space can be graphically represented using the conventional x, y, and z axes. Separation of a three-parameter space into regions requires a dividing *plane*, or a curved *surface*.

graph on the side of each axis, both parameters have a large overlap between the target and nontarget distributions. In other words, each parameter, taken individually, is a poor predictor of cancer. Combining the two parameters with simple logic functions would only provide a small improvement. This is especially interesting since the two parameters contain information to *perfectly* separate the targets from the nontargets. This is done by drawing a diagonal line between the two groups, as shown in the figure.

In the jargon of the field, this type of coordinate system is called a **parameter space**. For example, the two-dimensional plane in this example could be called a diameter-brightness space. The idea is that targets will occupy one region of the parameter space, while nontargets will occupy another. Separation between the two regions may be as simple as a straight line, or as complicated as closed regions with irregular borders. Figure 26-4 shows the next level of complexity, a three-parameter space being represented on the x, y and z axes. For example, this might correspond to a cancer detection system that measures *diameter*, *brightness*, and some third parameter, say, *edge sharpness*. Just as in the two-dimensional case, the important idea is that the members of the target and nontarget groups will (hopefully) occupy different regions of the space, allowing the two to be separated. In three dimensions, regions are separated by planes and curved surfaces. The term **hyperspace** (over, above, or beyond normal space) is often used to describe parameter spaces with more than three dimensions. Mathematically, hyperspaces are no different from one, two and three-dimensional spaces; however, they have the practical problem of not being able to be displayed in a graphical form in our three-dimensional universe.

The threshold selected for a single parameter problem cannot (usually) be classified as right or wrong. This is because each threshold value results in a unique combination of false-positives and false-negatives, i.e., some point along the ROC curve. This is trading one goal for another, and has no absolutely correct answer. On the other hand, parameter spaces with two or